

課程資訊	
課程名稱	數位控制系統 DIGITAL CONTROL SYSTEM
開課學期	95-2
授課對象	工學院 機械工程學研究所
授課教師	陳明新
課號	ME5247
課程識別碼	522 U5350
班次	
學分	3
全/半年	半年
必/選修	選修
上課時間	星期四6,7,8(13:20~16:20)
上課地點	普405
備註	系統控制知識領域 總人數上限：80人
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖
課程大綱	
為確保您的權利,請尊重智慧財產權及不得非法影印	
課程概述	Since almost every engineering branch (control, signal processing, communications, dynamic system analysis, ...) uses computers as their tools to handle the information, knowledge of digital system (or discrete-time system) become fundamental for every engineer. This course will introduce the basic skills required for the analysis and control of such systems. It is a course suitable for students who have taken the Automatic Control course. This course is mainly of undergraduate level, but is also suitable for graduate students who are not familiar with digital control.
課程目標	The main topics of this course are: 1. Z-transform 2. discrete-time model of sampled system 3. sampling theorem 4. state-space control 5. discrete transfer function control 6. optimal control and optimal observer
課程要求	The main topics of this course are: 1. Z-transform 2. discrete-time model of sampled system

	3. sampling theorem 4. state-space control 5. discrete transfer function control 6. optimal control and optimal observer	
預期每週課前 或/與課後學 習時數		
Office Hours		
指定閱讀		
參考書目	[1] Digital control of dynamic systems, by Frankin, Powell, and Workman, Addison Wesley, 1998. [2] Computer-controlled systems, theory and design. 3rd Ed., by K.J. Astrom, and B. Wittenmark, Prentice-Hall, 1997. [3] Control system design, by Goodwin, Graebe, Salgado, Prentice-Hall, 2001.	
評量方式 (僅供參考)	1. 本校尚無訂定 A+ 比例上限。 2. 本校採用等第制評定成績，學生成績評量辦法中的百分制分數區間與單科成績對照表 僅供參考，授課教師可依等第定義調整分數區間。詳見學習評量專區 (連結)。	
課程進度		
週次	日期	單元主題
無資料		